

Exchange Rate Regimes, Trade, and the Wage Comovements*

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Abstract

We construct a dynamic Ricardian model of trade with money and nominal exchange rate. The model implies stronger positive nominal wage comovements between trading countries that fix their bilateral exchange rates. Panel regression results based on data from OECD countries from 1973 to 2010 suggest that countries in the European Monetary Union (EMU) experienced stronger positive wage comovements with their main trade partners. In comparison, the positive wage comovements between countries engaged in non-currency-union pegs were weaker. When we restrict the regression to the subsample of the EMU countries, we find a significant increase in wage comovements after these countries joined the EMU in 1999 compared to the pre-euro era. Time series analysis based on cointegration tests also provides supporting evidence for the prediction of the model.

JEL classification: F1, F3

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1 Introduction

Typical Ricardian models of trade are real models and thus do not explicitly deal with the monetary aspects of trade such as changes in nominal prices and exchange rate, albeit Dornbusch et al. (1977) and Ito and Ohyama (1985) point out that it is possible to extend the discussion of the standard Ricardian model to deal with the monetary aspects of trade. In this paper, we construct a dynamic Ricardian model with money and nominal exchange rate and obtain an interesting theoretical prediction regarding the effects of exchange rate regimes and trade on the comovements of nominal wages: if a country fixes the exchange rate with its main trade partner, then its wage will comove strongly and positively with its partner's wage to preserve stability in relative wages and prices.¹ Intuitively, a fixed exchange rate implies that the two countries have similar growth rates in money supplies that then lead to similar growth rates in nominal wages. However, if a country floats the exchange rate, then its wage does not necessarily comove with that of its main trade partner, as the exchange rate can adjust to maintain the relative wages and prices. Thus the model predicts stronger comovements of wages between countries that fix their bilateral exchange rates.

In practice, many countries adopt fixed exchange rate regimes, with the extreme case being a currency union. In the 1990s, among the 91 economies studied by Sterne (1999), the number of countries adopting an explicit exchange rate target increased from 30 to 47. In 1999, the creation of the European Monetary Union (EMU) locked in 11 European countries committed to a single currency, and the EMU has been expanding since. However, in the wake of recent crises in peripheral countries in the EMU, economic commentators (for instance Economist (2010)), suggest that relative to Germany, countries such as Greece and Ireland have wages that are too high for their products to be competitive internationally. Yet, as EMU members, they do not have the option of

¹Our interests are in the comovements of nominal wages, so in the following discussion the word “wage(s)” refers to nominal wage(s), unless otherwise noted.

devaluation to promote their products. Such observations suggest that for countries in a currency union and countries adopting currency pegs, whether their wages align with those of their main trade partners has important economic consequences.

To the best of our knowledge, however, no previous studies have examined theoretically or empirically the effects of exchange rate regimes and trade on wage comovements between countries. Our paper now fills this void by constructing a model of exchange rate, trade, and wage comovements, and by testing the model's central prediction that wages comove strongly and positively between countries that peg their currencies. The results of panel regressions based on data from 24 OECD countries from 1973 to 2010 suggest that if a country and its main trade partner were in the EMU, then their wages experienced stronger comovements, especially in the longer term. However, for countries engaged in non-currency-union pegs, the positive wage comovements were weaker. We also run regressions with the sample restricted to EMU countries to determine whether joining the EMU in 1999 was associated with stronger positive wage comovements. We find that for EMU countries, there was a significant increase in wage comovements after joining the EMU, compared to the pre-euro era.

In addition, we implement the Johansen cointegration test to verify the prediction in terms of time-series that if two countries fix their bilateral exchange rate, their wages will be cointegrated. We find that out of the 11 EMU member countries whose main trade partners were also in the EMU, 5 countries experienced wage cointegration with the trade partners after the formation of the EMU in 1999. Meanwhile, for countries other than the 11 EMU members, the period after 1999 did not provide more evidence of wage cointegration. Hence, the time-series evidence also suggests that the EMU enhanced wage comovements.

Many previous studies analyze the comovements of wages with other variables, such as output, prices, and (un)employment. Surprisingly, however, there are few studies that

analyze the comovements of wages. In fact, as far as we are aware, there are only three studies on the subject.

The first, Budd, Konings and Slaughter (2002), highlights the comovements of wages within a multinational. They show the existence of comovements of wages within a multinational firm through internal risk sharing. The second, Robertson (2000), highlights the comovements of wages between the interior and border regions in a country. He provides evidence for these comovements between the interior and border regions of Mexico, thus indicating that the wage impact of emigration is transmitted to the overall Mexican economy. The third, Lamo, Perez and Schuknecht (2008), highlights the comovements of wages across sectors within a country. They show strong positive comovements of public and private sector wages over business cycles since the 1960s in the euro area and a number of other OECD countries.²

Our paper thus makes the following contributions to both the trade literature and the wage literature. First, because in reality countries often use different currencies, both exchange rates and nominal wages are important determinants of the relative prices of exports to imports. Past trade studies, however, have paid little attention to the monetary aspects of trade. By incorporating the monetary aspects of trade into the Ricardian model and then empirically testing the resulting implications, we reveal the difference in the wage effects of trade under the fixed and floating exchange rate regimes.

Second, we highlight that for countries engage extensively in trade, the choice of a fixed exchange rate regime will enhance wage comovements between countries. Our results add to the knowledge of wage comovements, as previous work has focused on inter- and intra-country wage comovements due to internal risk sharing within a multinational firm and emigration.

²Lamo et al. (2008) also study causal linkages between public and private sector wages, i.e. the public/private wage leadership. Their causality analysis suggests that although influences from the private sector appear on the whole to be stronger, there are direct and indirect feedback effects from public wage setting in a number of countries as well. See the references in their paper for studies on wage leadership in a particular country (mainly Sweden plus a few others).

Third, our results add to the knowledge of how an exchange rate peg acts as a nominal anchor. A currency peg or membership of a monetary union is one way to provide a nominal anchor for a country's output prices or inflation rate (Edwards, 1993; Calvo and Vegh, 1994; Willett, 1998). Our results suggest that in addition to providing a nominal anchor for output prices or the inflation rate, a monetary union can provide a nominal anchor for wages.

The rest of this paper is organized as follows. In Section 2, we construct a dynamic Ricardian model with money and nominal exchange rate and derive the predictions on wage comovements. We document the regression specification and data in Section 3. Section 4 reports regression results and cointegration analysis. We offer a discussion of the results in Section 5 and conclude in Section 6.

2 Theory

In this section, we construct a dynamic Ricardian model of trade with money and nominal exchange rate and thus address the monetary aspects of trade. The setup of the model borrows from two main sources, first, the Ricardian models (Dornbusch et al., 1977; Eaton and Kortum, 2002; Levchenko and Zhang, 2011), and second, models of money and exchange rates (Chari, Kehoe and McGrattan, 2002). The model in our paper is highly stylized, but it allows us to obtain analytic solutions, and insights about the effects of exchange rate regimes and trade on wage comovements.

2.1 Model setup

There are two countries, home and foreign. The variables associated with the foreign country are indicated by a superscript $*$. Each country is capable of producing the same continuum of traded goods. The traded goods are indexed by i , and $i \in [0, 1]$. Each country also produces a non-traded good, respectively.

The period preference for home country's representative consumer is (note the time

subscript t is suppressed, wherever possible)

$$U_t = \ln(C_t) + \chi \ln\left(\frac{M_t}{P_t}\right) + \kappa \frac{(1 - L_t)^{1-\gamma}}{1 - \gamma}$$

where

$$C_t = \left[\left(\int_0^1 C_{it}^{\frac{\eta-1}{\eta}} di \right)^{\epsilon \cdot \frac{\eta}{\eta-1}} \cdot C_{zt}^{1-\epsilon} \right]$$

and the period budget constraint is

$$\int P_{it} C_{it} di + P_{zt} C_{zt} + M_t + \sum_s q_{t+1}(s_{t+1}) B_{t+1}(s_{t+1}) = W_t L_t + M_{t-1} + B_t + \Pi_t + T_t \quad (1)$$

Here, C_i denotes the consumption of traded good i in the home country, and C_z is the consumption of the non-traded good in the home country. The variables M and P are money supply and the aggregate price level. The quantity L is labor in the home country, and the normalized time endowment of the home household is 1. The variables P_i , P_z , and W are the nominal price of traded good i , the nominal price of non-traded good, and nominal wage. The variables $q_{t+1}(s_{t+1})$ and $B_{t+1}(s_{t+1})$ are the price and the quantity of a nominal bond that pays one unit of home currency in state s in period $t + 1$, and zero otherwise. Lastly, Π and T are profits of home firms and a nominal transfer from the home government. Because our focus is on wages, we drop physical capital for the basic model, as in Eaton and Kortum (2002).

The production technology in the home country is

$$Y_{it} = \exp(u_t) A_{it} L_{it} \quad (2)$$

$$Y_{zt} = \exp(v_t) A_{zt} L_{zt} \quad (3)$$

where A_i is the productivity of good i . The random variable u_t is the productivity shock for all the traded goods. The random variable v_t is the productivity shock for the non-traded goods. The shocks are iid shocks with zero means. The market for each traded

good is perfectly competitive. The home producers of good i will have to compete with the foreign producer of firm i . Home and foreign consumers only buy from the producers with the lowest price, and there is no iceberg shipping cost.

Given the linear production technology and perfect competition, the local-currency prices charged by home and foreign producers of good i are

$$P_{it}^H = \frac{W_t}{\exp(u_t)A_i} \quad (4)$$

$$P_{it}^F = \frac{W_t^*}{\exp(u_t^*)A_i^*} \quad (5)$$

The prevailing home price for good i is

$$P_{it} = \min \{P_{it}^H, P_{it}^F e_t\}$$

where e_t is the nominal exchange rate, defined as the price of foreign currency in home currency. We order the varieties i such that A_i/A_i^* is decreasing in i . As in Dornbusch et al. (1977), we assume that A_i/A_i^* is strictly decreasing in i . Given u_t , and u_t^* , let $k \in [0, 1]$ be the cutoff variety for which $P_{kt}^H = e_t * P_{kt}^F$. Then the home country will produce all varieties with indices $i < k$, and the foreign country will produce all variety with $i > k$. We assume that both countries produce good k and there is no international trade of the variety k . The market for non-traded goods is also perfectly competitive. Consequently, the local-currency prices for the nontradable goods are

$$P_{zt} = \frac{W_t}{\exp(v_t)A_z} \quad (6)$$

$$P_{zt}^* = \frac{W_t^*}{\exp(v_t^*)A_z^*} \quad (7)$$

As in Chari et al. (2002), money is introduced into the utility function, and the real exchange rate is determined by the ratio of home and foreign marginal utility of consumption. The money supplies in the two countries follow stochastic processes, to be specified in later subsections for different exchange rate regimes. Any new money balance

$M_t - M_{t-1}$ is distributed to households through lumpsum transfer. That is, $T_t = M_t - M_{t-1}$. Because our focus is not on nominal price rigidity, we assume flexible prices.

At the beginning of each period, both monetary shocks and productivity shocks are observed by all players in the economy. Then firms post prices, consumers make purchase decisions, production and trade occur, and markets clear. The market clearing conditions are

$$\int_0^k L_{it} di + L_{zt} = L_t \quad (8)$$

$$\int_k^1 L_{it}^* di + L_{zt}^* = L_t^* \quad (9)$$

$$Y_{it} = C_{it} + C_{it}^* \quad \forall i < k \quad (10)$$

$$Y_{it}^* = C_{it} + C_{it}^* \quad \forall i > k \quad (11)$$

$$Y_{zt} = C_{zt}, Y_{kt}^* = C_{zt}^* \quad (12)$$

In addition, international trade is balanced:

$$\int_0^k C_{it}^* P_{it} di = \int_k^1 C_{it} P_{it} di \quad (13)$$

2.2 Characterizing the solution

For the household's maximization problem, the first order conditions with respect to C_{it} , C_{zt} , L_t , M_t , and $B_{t+1}(s)$ are

$$\frac{\epsilon(C_{it})^{\frac{-1}{\eta}}}{\int_0^1 (C_{it})^{\frac{\eta-1}{\eta}} di} = P_{it} \lambda_t \quad (14)$$

$$\frac{1 - \epsilon}{C_{zt}} = P_{zt} \lambda_t \quad (15)$$

$$\kappa(1 - L_t)^{-\gamma} = W_t \lambda_t \quad (16)$$

$$\frac{\chi}{M_t} = \lambda_t - \beta E_t \lambda_{t+1} \quad (17)$$

$$\beta E_t \lambda_{t+1}(s_{t+1}) = q_{t+1}(s_{t+1}) \cdot \lambda_t \quad (18)$$

The equilibrium exchange rate follows from the argument of (Chari et al., 2002)³

$$e_t = \frac{C_{zt}P_{zt}}{C_{zt}^*P_{zt}^*}\delta \quad (19)$$

where δ is a constant determined by initial conditions in period 0

$$\delta = \frac{C_{z0}^*P_{z0}^*}{C_{z0}P_{z0}}e_0 \quad (20)$$

Before we establish the link between exchange rate and nominal wages, we first show that the labor supply is constant in this model, a result useful in obtaining analytic solutions for wages.

Lemma 1. *The home and foreign labor supplies, L_t and L_t^* , are constants in all periods.*

Proof: Note equations (3), (12), (6), and (16) imply

$$\begin{aligned} C_{zt} &= L_{zt}A_z \exp(v_t) \\ P_{zt} &= \frac{W_t}{A_z \exp(v_t)} \\ \lambda_t &= \frac{\kappa(1 - L_t)^{-\gamma}}{W_t} \end{aligned}$$

Use these equations to eliminate C_{zt} , P_{zt} , and λ_t from equation (15), we have

$$1 - \epsilon = L_{zt}(1 - L_t)^{-\gamma} \quad (21)$$

Rewrite (14) as

$$\frac{\epsilon(C_{it})^{\frac{\eta-1}{\eta}}}{\int_0^1 (C_{it})^{\frac{\eta-1}{\eta}} di} = C_{it}P_{it}\lambda_t$$

Integrate both sides with respect to C_{it} , we have

$$\begin{aligned} \int_0^1 C_{it}P_{it}di &= \frac{\epsilon}{\lambda_t} \Rightarrow \\ \int_0^k C_{it}P_{it}di + \int_k^1 C_{it}P_{it}di &= \frac{\epsilon}{\lambda_t} \end{aligned}$$

³See equation (14) in page 544 of Chari et al. (2002)

Substitute in the balanced trade condition, we have

$$\begin{aligned} \int_0^k C_{it} P_{it} di + \int_0^k C_{it}^* P_{it} di &= \frac{\epsilon}{\lambda_t} \Rightarrow \\ \int_0^k (C_{it} + C_{it}^*) P_{it} di &= \frac{\epsilon}{\lambda_t} \end{aligned} \quad (22)$$

Note equations (4), (2), and (10) imply

$$\begin{aligned} C_{it} + C_{it}^* &= L_{it} A_i \cdot \exp(u_t) \\ P_{it} &= \frac{W_t}{A_i \cdot \exp(u_t)} \quad \forall i < k \end{aligned}$$

Substitute these equations into equation (22), we obtain

$$\int_0^k L_{it} di = \frac{\epsilon}{W_t \lambda_t} \quad (23)$$

Similarly, combining equations (3), (12), (6), (15) and (16), we can obtain

$$L_{zt} = \frac{1 - \epsilon}{W_t \lambda_t} \quad (24)$$

Combining (23) and (24) yields

$$\frac{\int_0^k L_{it} di}{L_{zt}} = \frac{\epsilon}{1 - \epsilon} \quad (25)$$

Use equation (25) to eliminate $\int_0^k L_{it} di$ from equation (8), we have

$$L_{zt} = (1 - \epsilon) L_t$$

Substitute the last line into equation (21), we can obtain

$$L_t + L_t^{1/\gamma} = 1 \quad (26)$$

Note that the left hand side is a strictly increasing function on the support of 0 to 1. The left hand side takes the value of 0 when $L_t = 0$ and the value of 2 when $L_t = 1$. Therefore, there is a unique value of L_t that solves the equation (26). This implies the labor supply is constant in all periods. The proof for L_t^* is similar. ■

We now establish the link between exchange rate and nominal wages. From equations (15) and (16), we know

$$\begin{aligned}\frac{W_t}{W_t^*} &= \frac{\kappa(1-L_t)^{-\gamma}/\lambda_t}{\kappa(1-L_t^*)^{-\gamma}/\lambda_t^*} \\ &= \frac{(1-L_t)^{-\gamma}}{(1-L_t^*)^{-\gamma}} \cdot \frac{(1-\epsilon)/C_{zt}^*P_{zt}^*}{(1-\epsilon)/C_{zt}P_{zt}} \\ &= \frac{(1-L_t)^{-\gamma}}{(1-L_t^*)^{-\gamma}} \cdot \frac{C_{zt}P_{zt}}{C_{zt}^*P_{zt}^*}\end{aligned}$$

Use (19), we have

$$\frac{W_t}{W_t^*} = \frac{(1-L_t)^{-\gamma}}{(1-L_t^*)^{-\gamma}} \frac{e_t}{\delta} \quad (27)$$

Note from Lemma 1, we know L_t and L_t^* are both constants. Then from (27), it is obvious that if the exchange rate is fixed, the home wage must move 1-to-1 with the foreign wage.

Note that the relative wage satisfies the balance of trade in terms of labor:

$$\begin{aligned}\int_0^k C_{it}^* P_{it} di &= \int_k^1 C_{it} P_{it} di \\ \int_0^k C_{it}^* P_{it}^H di &= \int_k^1 C_{it} P_{it}^F e_t di \\ \int_0^k C_{it}^* \frac{W_t}{\exp(u_t)A_i} di &= \int_k^1 C_{it} \frac{W_t^* e_t}{\exp(u_t^*)A_z^*} di\end{aligned}$$

The above equation means that the balance of trade in terms of labor is balanced because $\int_0^k \frac{C_{it}^*}{\exp(u_t)A_i} di$ is the amount of home labor devoted to the production of home export goods and $\int_k^1 \frac{C_{it}}{\exp(u_t^*)A_z^*} di$ is the amount of foreign labor devoted to the production of foreign export goods. From this balance of trade in terms of labor, we obtain the relative wage as:

$$\frac{W_t}{W_t^* e_t} = \frac{\int_k^1 C_{it} \frac{1}{\exp(u_t^*)A_z^*} di}{\int_0^k C_{it}^* \frac{1}{\exp(u_t)A_i} di}.$$

2.3 Wage comovements as functions of monetary and productivity shocks under a fixed exchange rate regime

Assume that the foreign money supply follows the stochastic process

$$M_t^* = \exp(\mu_t^*) M_{t-1}^* (1 + g) \quad (28)$$

The constant g is the growth rate of money. We assume that μ_t^* are zero-mean iid shocks with cdf $\Phi(\mu)$. The home country's monetary authority would set M_t to fix the exchange rate such that $e_t = e_{t+1}$.

Proposition 1. *Under the fixed exchange rate regime, there are perfect comovements between home and foreign nominal wages. Specifically, the changes in wages in both countries are determined by the monetary shocks in the foreign country, $W_t/W_{t-1} = W_t^*/W_{t-1}^* = \exp(\mu_t^*)$.*

Proof: From the foreign version of equation (17), we have

$$\begin{aligned} \lambda_t^* &= \frac{\chi}{M_t^*} + \beta E_t \lambda_{t+1}^* \\ &= \frac{\chi}{M_t^*} + \beta E_t \left(\frac{\chi}{M_{t+1}^*} + \beta E_{t+1} \lambda_{t+2}^* \right) \\ &= \frac{\chi}{M_t^*} + \beta E_t \left(\frac{\chi}{M_{t+1}^*} \right) + \beta^2 E_t \left(\frac{\chi}{M_{t+2}^*} \right) + \beta^3 E_t \left(\frac{\chi}{M_{t+3}^*} \right) + \dots \\ &= \frac{\chi}{M_t^*} + \beta E_t \left(\frac{\chi}{M_t^* \exp(\mu_{t+1}^*) (1 + g)} \right) + \beta^2 E_t \left(\frac{\chi}{M_t^* \exp(\mu_{t+1}^*) \exp(\mu_{t+2}^*) (1 + g)^2} \right) + \\ &\quad \beta^3 E_t \left(\frac{\chi}{M_t^* \exp(\mu_{t+1}^*) \exp(\mu_{t+2}^*) \exp(\mu_{t+3}^*) (1 + g)^3} \right) + \dots \\ &= \frac{\chi}{M_t^*} \left[1 + \frac{\beta}{1 + g} E_t \left(\frac{1}{\exp(\mu_{t+1}^*)} \right) + \left(\frac{\beta}{1 + g} \right)^2 E_t \left(\frac{1}{\exp(\mu_{t+1}^*) \exp(\mu_{t+2}^*)} \right) \right. \\ &\quad \left. + \left(\frac{\beta}{1 + g} \right)^3 E_t \left(\frac{1}{\exp(\mu_{t+1}^*) \exp(\mu_{t+2}^*) \exp(\mu_{t+3}^*)} \right) + \dots \right] \\ &= \frac{\chi}{M_t^*} \left[1 + \frac{\beta}{1 + g} \int \frac{1}{\exp(\mu^*)} d\Phi(\mu^*) + \left(\frac{\beta}{1 + g} \right)^2 \int \int \frac{1}{\exp(\mu^*)} d\Phi(\mu^*) \frac{1}{\exp(\mu^*)} d\Phi(\mu^*) \right. \\ &\quad \left. + \left(\frac{\beta}{1 + g} \right)^3 \int \int \int \frac{1}{\exp(\mu^*)} d\Phi(\mu^*) \frac{1}{\exp(\mu^*)} d\Phi(\mu^*) \frac{1}{\exp(\mu^*)} d\Phi(\mu^*) + \dots \right] \end{aligned}$$

The terms in the brackets are a constant. Define the constant as ψ , we have

$$\lambda_t^* = \frac{\chi\psi}{M_t^*} = \frac{\chi\psi}{M_{t-1}^* \exp(\mu_t^*)(1+g)} \quad (29)$$

Substitute equation (29) into the foreign version of (16), we have

$$\begin{aligned} W_t^* &= \kappa(1 - L_t^*)^{-\gamma} \frac{M_{t-1}^* \exp(\mu_t^*)(1+g)\gamma}{\chi\psi} \\ &= \kappa(1 - L_{t-1}^*)^{-\gamma} \frac{M_{t-1}^* \exp(\mu_t^*)(1+g)\gamma}{\chi\psi} \\ &= W_{t-1}^* \exp(\mu_t^*)(1+g) \end{aligned} \quad (30)$$

where the second equality follows from Lemma 1. From equation (27), we can see that to fix the exchange rate, it must hold that

$$\frac{W_t}{W_t^*} = \frac{(1 - L_t)^{-\gamma} e_t}{(1 - L_t^*)^{-\gamma} \delta} = \frac{(1 - L_{t-1})^{-\gamma} e_{t-1}}{(1 - L_{t-1}^*)^{-\gamma} \delta} = \frac{W_{t-1}}{W_{t-1}^*}$$

Using (30), we obtain

$$W_t = W_{t-1} \exp(\mu_t^*)(1+g) \quad (31)$$

From equations (31) and (30), it is obvious that the wage comovement is perfect because both changes in W_t and W_t^* are driven by the monetary shock in the foreign country. ■

Note, to fix the exchange rate the home country has to match the change in foreign money supply. That is, $M_t = M_{t-1} \exp(\mu_t^*)(1+g)$.

Proposition 2. *Shocks to productivities of tradable goods only affect the amounts of labor used in producing export goods in the two countries, not the relative wage $W_t/(W_t^* e_t)$.*

Proof: The nominal prices for tradable goods, in home currency, are

$$\begin{aligned} P_{it} &= \frac{W_t}{\exp(u_t) A_i} \quad \forall i < k \\ P_{it} &= \frac{W_t^* e_t}{\exp(u_t^*) A_i^*} \quad \forall i \geq k \end{aligned}$$

Substitute the prices into the balanced-trade condition (13), we have

$$\int_0^k C_{it}^* \frac{W_t}{\exp(u_t)A_i} di = \int_k^1 C_{it} \frac{W_t^* e_t}{\exp(u_t^*)A_i^*} di$$

Rearrangement yields

$$\frac{W_t}{W_t^* e_t} \frac{\int_0^k \frac{C_{it}^*}{A_i} di}{\int_k^1 \frac{C_{it}}{A_i^*} di} = \frac{\exp(u_t)}{\exp(u_t^*)} \quad (32)$$

Note that $\int_0^k \frac{C_{it}^*}{A_i} di$ is the amount of home labor used in producing home export goods and $\int_k^1 \frac{C_{it}}{A_i^*} di$ is the amount of foreign labor used in producing foreign export goods. Meanwhile, equation (27) implies the relative wage $\frac{W_t}{W_t^* e_t} = \frac{(1-L_t)^{-\gamma}}{(1-L_t^*)^{-\gamma}} \frac{1}{\delta}$ is constant. Hence, the last equation becomes

$$\frac{(1-L_t)^{-\gamma}}{(1-L_t^*)^{-\gamma}} \frac{1}{\delta} \frac{\int_0^k \frac{C_{it}^*}{A_i} di}{\int_k^1 \frac{C_{it}}{A_i^*} di} = \frac{\exp(u_t)}{\exp(u_t^*)} \quad (33)$$

From equation (33), it is obvious that a positive foreign productivity shock (relative to the home productivity shock) will result in a decline in home labor used in producing home export goods (relative to the foreign labor used in producing foreign export goods). ■

An alternative way to interpret (32) is to rewrite it as

$$\frac{W_t}{W_t^* e_t} \int_0^k \frac{C_{it}^*}{\exp(u_t)A_i} di = \int_k^1 \frac{C_{it}}{\exp(u_t^*)A_i^*} di$$

Hence, the equation says that the amounts of labor devoted to the production of home export goods and foreign export goods remain proportional.

2.4 Wage comovements as functions of monetary and productivity shocks under a flexible exchange rate regime

Assume that the home and foreign money supplies follow the stochastic process

$$M_t = \exp(\mu_t) M_{t-1} (1 + g) \quad (34)$$

$$M_t^* = \exp(\mu_t^*) M_{t-1}^* (1 + g^*) \quad (35)$$

We assume that μ_t and μ_t^* are zero-mean iid shocks with a common cdf $\Phi(\mu)$.

Proposition 3. *Under the flexible exchange rate regime, unless the home and foreign monetary shocks are correlated, there is no comovements between home and foreign nominal wages.*

Proof: From the last subsection, we know

$$W_t = W_{t-1} \exp(\mu_t)(1 + g)$$

$$W_t^* = W_{t-1}^* \exp(\mu_t^*)(1 + g^*)$$

or

$$\ln(W_t) - \ln(W_{t-1}) = \mu_t + \ln(1 + g) \quad (36)$$

$$\ln(W_t^*) - \ln(W_{t-1}^*) = \mu_t^* + \ln(1 + g^*) \quad (37)$$

Consequently, only when home and foreign monetary shocks are positively correlated, the home and foreign wages will have positive comovement. ■

Proposition 4. *Shocks to productivities of tradable goods only affect the amounts of labor used in producing export goods in the two countries, not the relative wage $W_t/(W_t^* e_t)$.*

Proof: Similar to the proof of proposition 2. ■

Proposition 5. *Exchange rate determination. The equilibrium exchange rate is determined by the money supplies, i.e. $e_t = \delta M_t/M_t^*$.*

Proof: From equation (27) we know

$$e_t = \delta \frac{W_t (1 - L_t^*)^{-\gamma}}{W_t^* (1 - L_t)^{-\gamma}}$$

From the derivation of equation (30), we know

$$W_t = \kappa (1 - L_t)^{-\gamma} \frac{M_t \gamma}{\chi \psi}$$

$$W_t^* = \kappa (1 - L_t^*)^{-\gamma} \frac{M_t^* \gamma}{\chi \psi}$$

Substitute the last two equations into the expression for exchange rate above, we have

$$e_t = \delta \frac{M_t}{M_t^*}.$$

■

Note,

$$\frac{e_t}{e_{t-1}} = \frac{M_t}{M_{t-1}} \frac{M_{t-1}^*}{M_t^*} = \frac{\exp(\mu_t)(1+g)}{\exp(\mu_t^*)(1+g^*)}$$

In this section, we construct a dynamic Ricardian model with money and nominal exchange rate. The model implies that if the exchange rate with a trade partner is fixed, then the wage in the home country will have stronger positive comovements with that in the trade partner. On the other hand, if the exchange rate with the trade partner is floating, then the wage in the home country does not necessarily comove with that in the trade partner due to the flexibility in the exchange rate. The key mechanism is the following. In each country, a change in nominal wage is linked directly to the change in money supply. When the exchange rate is fixed, the growth rates in money supplies must be equal in the two countries, implying a common growth rate in nominal wages in the two countries. Meanwhile, when the exchange rate is flexible, money growth rates and wage growth rates in the two countries are not correlated in general.⁴

3 Regression Specification and Data

To test empirically whether wages would exhibit stronger positive comovements for countries that peg their currencies, we first estimate the following two baseline models with

⁴Note that the key predictions developed here also hold in the framework of the standard static Ricardian model, and in the standard Heckscher-Ohlin (H-O) model of trade. For the argument in a static Ricardian model, see the earlier version of this paper citeKPT2011. In the standard H-O model, by incorporating a nominal exchange rate and using the factor price equalization theorem, we obtain $W = eW^*$ and $R = eR^*$, where R and R^* are the home and foreign rentals, respectively. Thus it is obvious that under the fixed exchange rate regime, the home and foreign wages W and W^* would comove strongly and positively.

country fixed effects:

$$\begin{aligned}\Delta \ln(W_{it}) = & \beta_0 + \beta_1 \Delta \ln(W_{it}^*) \\ & + \beta_2 \cdot peg_{it} \cdot \Delta \ln(W_{it}^*) + \beta_3 \cdot trend + u_i + \epsilon_{it}\end{aligned}\quad (38)$$

$$\begin{aligned}\Delta \ln(W_{it}) = & \gamma_0 + \gamma_1 \Delta \ln(W_{it}^*) \\ & + \gamma_2 \cdot non\ CU\ peg_{it} \cdot \Delta \ln(W_{it}^*) \\ & + \gamma_3 \cdot CU_{it} \cdot \Delta \ln(W_{it}^*) + \gamma_4 \cdot trend + u_i + \epsilon_{it}\end{aligned}\quad (39)$$

where W_{it} is the wage in country i in year t , and W_{it}^* is the wage in country i 's base country in period t . Because the model generates predictions based on the nominal wages and exchange rates, we use only the nominal wage as the dependent variable. In specification 1 (equation (38)), peg_{it} is a dummy variable that takes the value 1 if country i 's currency is pegged to its base country in period t , and 0 otherwise. In specification 2 (equation (39)), we differentiate between two types of pegs. CU_{it} is a dummy variable indicating that country i and its base country are in a currency union in period t , and $non\ CU\ peg_{it}$ is a dummy variable indicating other types of pegs. The variable $trend$ is a linear time trend, and u_i is the country fixed effect.

The fixed effects in the specifications account for the time-constant idiosyncrasy in the growth in nominal wage, such as country-specific productivity path or inflation in a specific country relative to its base country. We also include a linear time trend to capture the trend in the growth rate of the nominal wages. Such a trend could arise due to the productivity slowdown since the 1970s and the decline in inflation rates in some countries.

If β_2 , γ_2 and γ_3 are positive, then the prediction of the modified Ricardian model that wages comove positively and strongly under a fixed exchange rate regime is supported. Because the Ricardian model is a general equilibrium model, we treat both home and foreign wages as endogenous to the model. Hence, in testing whether β_2 , γ_2 and γ_3 are positive, we are examining whether there is a positive conditional correlation between

home and foreign wage growth under a fixed exchange rate regime. Our empirical work does not attempt, and should not be interpreted as, the identification of the effect of exogenous wage changes in the foreign country on the home wage.

Our empirical analysis uses wage data from the Source OECD (www.sourceoecd.org) which provides detailed wage information of OECD countries starting from 1973. Wage is measured by the index for nominal hourly earnings in manufacturing sectors. We made this choice because the products of the manufacturing sectors are highly tradable.

The classification of the exchange rate regime follows Klein and Shambaugh (2006), who determine whether a country pegs its currency to the base country, based on the volatility of the exchange rate. In Klein and Shambaugh (2006), country i 's base country is the country to which country i pegs its exchange rate or the country with which country i has the most significant trade relationship. We also follow their choice of base countries.⁵

Because we are looking at OECD countries, the currency union is the EMU. The countries included in our sample are Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hungary, Iceland, Italy, Japan, Korea, Luxembourg, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Spain, Sweden, Turkey, and the UK. The US is not in the sample because the US does not have a dominant trade-partner to be used as the base country. Our sample covers data from the first quarter of 1973 to the fourth quarter of 2010. The details about the base country, episodes of exchange rate pegs, and data range for each country are documented in Table 6.

4 Empirical Results

4.1 Main Regression Results

We run regressions with growth rates of wages calculated over different time intervals. This is because our model assumes perfect wage flexibility that is more likely to be true in the long run, and regressions over different time intervals will reveal whether the prediction of

⁵The description of their data can be found at <http://www.dartmouth.edu/~jshambau/>

the model is more accurate in longer time horizon. The top rows in Table 1 through Table 5 indicate the frequency at which we calculate the growth rates of the wage in country i and its base country. We choose to use the wage growth over a quarter, a year, two years, and four years.

The results in Table 1, corresponding to specification 1 (equation (38)), suggest that pegs have no significant effect on wage comovements at 5% significance level. This seems unsupportive to our hypothesis. In Table 2, we follow specification 2 (equation (39)) to differentiate between the two types of pegs. While the non-currency-union pegs reduce wage comovements at some frequencies, the EMU always increases wage comovements significantly. For instance, at the quarterly frequency, the coefficients on the interaction between the currency union and wage growth in the base country is 0.64. This result implies that if the wage in country i 's base country increases by 1%, being in a currency union with the base country predicts an additional increase of 0.64% in country i 's wage. Interestingly, at the two-year frequency, the coefficient on the interaction is 0.96, which is close to unity.

In addition, as indicated by the coefficient of wage growth in the base country ($\Delta \ln(W_t^*)$) in Tables 1 and 2, wages in general do comove positively between a country and its base country at the quarterly frequency. For instance, the coefficients are 0.27 and 0.23 for the quarterly data, which implies that when the wage in country i 's base country increases by 1%, the wage in country i increases by 0.27% and 0.23%, respectively.

By comparing results for different frequencies, we see that the R^2 increases with the length of the time interval. For instance, in Table 2, the R^2 associated with quarterly wage growth is 0.22. When we use the growth rate over four years, the R^2 increases to 0.49, indicating that our simple model appears to have good explanatory power with respect to wage growth in the long run.

Although our dynamic Ricardian model does have the productivity and monetary

shocks as possible factors affecting wage comovements, the above baseline specifications (38) and (39) do not include these two factors. The model indicates that productivity shocks cannot affect wage comovements while the synchronization of inflation rates is the central mechanism. To ask whether there are still positive wage comovements after controlling for productivity growth and whether inflation enhances wage comovements, we add labor productivity growth and CPI inflation of a country to the regression. Here, the measure for labor productivity is output per worker in the industrial sectors.⁶

The results associated with this modification are presented in Table 3 and Table 4. Comparing Table 3 to Table 1, we now see pegs in general no longer reduce the positive wage comovements between country pairs. In particular, for wage growth over two years, pegs in general strengthen the positive wage comovements, although the strengthening effects are weak statistically. Comparing Table 4 to Table 2, there are two differences. First, the interaction between the currency union and wage growth in the base country is positive and significant only when we look at growth over two years. Second, the interaction between non-currency-union pegs and wage growth in the base country no longer has a negative and significant coefficient. The coefficient is actually positive and marginally significant for the wage growth over two years. After controlling for productivity growth and inflation, these regression results seem to suggest that the positive effect of currency pegs on wage comovements appears mainly in the intermediate term over two-year periods. We will discuss this finding in more details in Section 5.

4.2 Regression Results Based on EMU Member Countries

In Table 5, we repeat the estimations in Table 4 but restrict the sample to countries currently in the EMU. More specifically, the countries included are Austria, Belgium, Finland, France, Germany, Ireland, Italy, Luxembourg, the Netherlands, Portugal, and

⁶The CPI data are obtained from the Source OECD. Labor productivity is computed as the constant-price output in the industry divided by employment in the industry. Both output and employment in the industry are also obtained from the Source OECD.

Spain. The time range remains 1973 to 2010. These estimations serve two purposes. First, we want to explore if non-currency-union pegs worked differently for countries not in the EMU or for countries that later joined the EMU. Second, the additional regressions provide direct evidence of whether joining the EMU is associated with stronger positive wage comovements.

Relative to Table 4, the results regarding the coefficients on non-currency-union pegs in Table 5 are very similar. Non-currency-pegs only strengthened the wage comovements when we focus on the wage growth over two-year intervals. We also run regressions with the countries not in the EMU and find that non-currency-pegs have no strengthening effect on wage comovements, except that the pegs have marginally positive effects when the dependent variable is the wage growth over two-years. To preserve space, we do not report these results in the paper. These regression results indicate that non-currency-pegs had similar effects on wage comovements for countries in and out of the EMU.

Meanwhile, regarding the coefficients on the EMU dummy-base country wage interaction in Table 5, the signs are all positive. Compared to Table 4, the significance level is higher for wage growth over two-year and four-year intervals. Hence, for the same 11 countries, the positive wage comovements with their base countries after joining the EMU are stronger than before joining the EMU.

4.3 Robustness Checks

In unreported regressions, we estimate another pair of alternative specifications by replacing the linear time trend with time fixed effects. The results are similar to Tables 1 and 2, but weaker. However, we believe that time fixed effects are not suitable for testing the theoretical predictions. To see the reason, note that Germany is the reference for almost all countries in the EMU. If our hypothesis is correct, wages in these EMU countries should comove with German wages in each time period. However, when we include the time fixed effects, the regressions will attribute such comovements with the German wages

to time-specific effects, rather than attributing them to the currency union, which is a time-constant fundamental.

Finally, we add the dummy variables for pegs in general, currency union, and non-currency-union pegs to the regressions that are not reported in this paper. These stand-alone dummy variables are almost always insignificant. The insignificance, consistent with the model, also makes sense intuitively, as there is no obvious reason why wage growth rates should be higher or lower in countries that fix their exchange rates.

4.4 Cointegration Analysis

In addition to the panel regressions, we also conduct the Johansen cointegration test (Johansen, 1991, 1995) for each country to see if its wage cointegrated with that in the base country.⁷ Note that equation (27) implies

$$W_t = \frac{(1 - L_t)^{-\gamma}}{(1 - L_t^*)^{-\gamma}} \frac{e_t}{\delta} W_t^* \quad (40)$$

When exchange rate e is fixed, then $W_t - \frac{(1-L_t)^{-\gamma}}{(1-L_t^*)^{-\gamma}} \frac{e_t}{\delta} W_t^*$ is stationary over time. Hence, if the bilateral exchange rate is fixed, W and W^* are likely to be cointegrated in the sense of Engle and Granger (1987). For countries that do not fix their bilateral exchange rate, their wages in general are not cointegrated. Therefore, equation (40) provides a prediction that can be tested with the Johansen cointegration test.

We summarize the test results in Table 6.⁸ The 24 countries in our sample are split into three groups. The first are the original EMU members in 1999. The second contains a sole country, Denmark, which is not in the EMU but pegs its currency to the Euro via

⁷If there exists a common long-run relationship (between home and base country wages in our case) among different countries, it is possible to use panel cointegration to pool information from different countries (Pedroni, 1999). However, since we believe that within our sample, the relationship between wages are heterogeneous due to the difference in exchange rate regime and other reasons, we choose not to use a panel cointegration test.

⁸Before implementing the Johansen test of cointegration, we conduct the Augmented Dickey-Fuller test (Dickey and Fuller, 1979) on each wage series. Since nominal wages rise with both inflation and productivity, it is not surprising that we find all wage series are non-stationary. Hence, it is not trivial to test for cointegration between two wage series.

the European Exchange Rate Mechanism II (ERM II). The third group contains all other countries. We implement the Johansen test for the EMU countries before and after the introduction of Euro in 1999. We do the same for all other countries, to check whether other global trends or events can contribute to cointegration of wages in all countries after 1999.

Columns 3 and 4 of Table 6 are the test results at 5% significance level.⁹ Three patterns emerge from the tests. First, for the EMU countries, there is no evidence of wage cointegration prior to 1999. After 1999, 5 of 11 countries exhibited wage cointegration with their base countries that were also in the EMU.¹⁰ Second, intermittent pegs adopted by all countries, including EMU members, appear to have no systematic effect on wage cointegration. Meanwhile, Denmark, who pegs to the Euro consistently after 1999 via ERM II, did exhibit wage cointegration with its base country, Germany. Third, there seems to be no global events around 1999 that induce wage cointegration in general. Among the third group of countries, while Canada and Sweden switched from no cointegration to cointegration after 1999, Korea switched from cointegration to no cointegration.

Overall, the tests provide time-series evidence supporting the theoretical prediction that wages are likely to cointegrate if two countries fix their bilateral exchange rate persistently.

5 Discussion

In subsection 4.1, after controlling for inflation and productivity growth, the wage comovements appear in two-year periods. Because our model is based on the assumption of flexible wages that is more reasonable for a longer time-horizon, it appears that the predictions of wage comovements manifest themselves over a longer period of time.

⁹The critical values are obtained from MacKinnon, Haug and Michelis (1999).

¹⁰Our sample includes 12 EMU members. However, the base country for Germany, the US, is not in the EMU.

Subsection 4.2 shows the regression results with the sample restricted to the EMU countries. We find that the coefficients on currency-union pegs are all positive and that the significance levels are higher, while the coefficients on non-currency-union pegs change little. This finding indicates that non-currency-union pegs did not work differently for the countries that have never been in the EMU and the countries that joined the EMU in 1999. It also indicates that for EMU countries, there was a significant increase in wage comovements after joining the EMU in 1999, compared to the pre-euro era.

To complement the panel regression analysis, we provide time-series evidence supporting our model in subsection 4.3. The panel regressions draw on both time series and cross-section variation to estimate the period-to-period comovements in wages. There we account for country heterogeneity with fixed effects. When we test for cointegration country by country in subsection 4.3, we further recognize the importance of country heterogeneity by using only the time-series variation. Instead of estimating the average period-to-period comovements in wages across countries and periods, the cointegration tests focus on whether for each country, there is overall wage comovements with its base country over two spans of time, i.e. the time before and after the creation of the EMU.

Overall, the empirical results support the predictions of the dynamic Ricardian model with nominal exchange rate. As indicated by Tables 3 and 4, augmenting equations (38) and (39) with productivity growth and inflation better aligns the empirical results with the theory, in the sense that both currency union and non-currency-union pegs strengthen the positive wage comovements between country pairs. Perhaps it is not surprising that non-currency-union pegs only weakly enhanced wage comovements. As Obstfeld and Rogoff (1995) suggest, pegs of this type often lack credibility. Historically, countries had been known to break their pegs and devalue when the prices of their products were not competitive internationally. If producers expect such devaluations, then there are smaller incentives to align wages to the base country.

Our results are relevant for the debate on whether the EMU is an optimum currency area. The existence of wage comovements suggests that although relative to the US, the EMU originally was less likely to meet the criteria for optimum currency area (Feenstra and Taylor, 2008, p.879), it may have enhanced the economic integration of its members via wage comovements.

Finally, this paper is also related to the literature on the relationship between the relative price of non-traded goods and the bilateral real exchange rate. In two recent papers on the subject, Betts and Kehoe (2006) and Betts and Kehoe (2008), one of the key findings is that for pairs of countries who trade intensively or maintain a stable bilateral real exchange rate, the relative price of non-traded goods has a stronger relationship with their bilateral real exchange rate. In the dynamic Ricardian model in our paper, a similar result holds. It can be shown that when the exchange rate is fixed, the real exchange rate is determined by the relative price of non-traded goods.¹¹

6 Conclusion

We construct a dynamic Ricardian model of trade with money and nominal exchange rate and obtain the prediction that two countries' nominal wages must exhibit strong and positive comovements if they fix the bilateral exchange rate. We use the data from 24 OECD countries between 1973 and 2010 to test this prediction. We find that if a country and its main trade partner were in the EMU, their wages experienced stronger comovements, especially over two-year intervals. For country pairs who engage in non-currency-union pegs, their wages had weaker comovements, in terms of both magnitude and statistical significance. When we restrict our attention to the EMU members, we also find a significant increase in wage comovements after they joined the EMU in 1999 compared to the period before 1999. Cointegration analysis also suggests that the EMU

¹¹The price index for aggregate consumption good in the home country is $P_t = [\epsilon^{-\epsilon}(1 - \epsilon)^{\epsilon-1}](\int_0^1 P_{it}^{1-\eta})^{\frac{1}{1-\eta}\epsilon}(P_{zt})^{1-\epsilon}$. Hence the real exchange rate is $re_t = e_t \frac{P_t^*}{P_t} = e_t^{1-\epsilon} (\frac{P_{zt}^*}{P_{zt}})^{1-\epsilon}$.

strengthened wage comovements while non-currency-union pegs lacked such strengthening effects.

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Table 1: Specification 1

Variables	Quarterly (1)	Annual (2)	2-year (3)	4-year (4)
$\Delta \ln(W_t^*)$	0.27 (0.13)**	0.23 (0.37)	0.3 (0.39)	0.28 (0.4)
$Peg_t \times \Delta \ln(W_t^*)$	-.23 (0.14)*	-.03 (0.16)	-.05 (0.16)	-.37 (0.27)
<i>Linear time trend</i>	-.0002 (0.0000455)***	-.003 (0.001)***	-.01 (0.004)***	-.05 (0.02)***
<i>Const.</i>	0.04 (0.005)***	0.14 (0.03)***	0.29 (0.07)***	0.62 (0.17)***
<i>Obs.</i>	2042	677	316	150
R^2	0.2	0.29	0.37	0.46
<i>F statistic</i>	60.1	56.43	48.51	46.48

Notes: (1) The dependent variable is the wage growth rate as measured by the log change in wages. (2) The variable $\Delta \ln(W_t^*)$ is the wage growth rate of the base country, and Peg_t is a dummy variable indicating whether a country pegs its exchange rate to its base country via a currency union or other arrangements. (3) The top row indicates the time interval at which the wage growth rates are calculated. (4) The numbers in the parentheses are robust standard errors. (5) *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. (6) All regressions include country fixed effects.

Table 2: Specification 2

Variables	Quarterly (1)	Annual (2)	2-year (3)	4-year (4)
$\Delta \ln(W_t^*)$	0.23 (0.13)*	0.15 (0.38)	0.23 (0.39)	0.16 (0.42)
$CU_t \times \Delta \ln(W_t^*)$	0.64 (0.18)***	0.78 (0.33)**	0.96 (0.34)***	0.68 (0.22)***
$Non\ CU\ peg_t \times \Delta \ln(W_t^*)$	-.35 (0.12)***	-.13 (0.15)	-.15 (0.17)	-.53 (0.23)**
<i>Linear time trend</i>	-.0002 (0.0000475)***	-.004 (0.001)***	-.02 (0.005)***	-.06 (0.02)***
<i>Const.</i>	0.04 (0.005)***	0.15 (0.03)***	0.31 (0.08)***	0.69 (0.18)***
<i>Obs.</i>	2042	677	316	150
R^2	0.22	0.3	0.39	0.49
<i>F statistic</i>	47.72	40.65	33.78	28.93

Notes: (1) The dependent variable is the wage growth rate as measured by the log change in wages. (2) The variable $\Delta \ln(W_t^*)$ is the wage growth rate of the base country, CU_t is a dummy variable indicating membership of a currency union, and $Non\ CU\ peg_t$ is a dummy variable indicating whether a country engages in a peg other than being a member of a currency union. (3) The top row indicates the time interval at which the wage growth rates are calculated. (4) The numbers in the parentheses are robust standard errors. (5) *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. (6) All regressions include country fixed effects.

Table 3: Specification 1, with productivity growth and CPI inflation

	Quarterly	Annual	2-year	4-year
Variables	(1)	(2)	(3)	(4)
$\Delta \ln(W_t^*)$	0.21 (0.08)***	0.27 (0.14)**	0.2 (0.18)	0.31 (0.19)*
$Peg_t \times \Delta \ln(W_t^*)$	-.13 (0.13)	0.02 (0.1)	0.18 (0.11)*	-.10 (0.17)
$\Delta \ln(a_t)$	-.06 (0.09)	0.02 (0.05)	0.02 (0.04)	0.12 (0.06)**
$CPI \text{ inflation rate}_t$	0.55 (0.15)***	0.77 (0.04)***	0.82 (0.03)***	0.8 (0.09)***
<i>Linear time trend</i>	-.0001 (0.0000387)***	-.0008 (0.0005)	-.003 (0.002)	-.007 (0.008)
<i>Const.</i>	0.02 (0.005)***	0.03 (0.02)**	0.06 (0.04)	0.1 (0.09)
<i>Obs.</i>	1899	622	295	139
R^2	0.38	0.72	0.77	0.84
<i>F statistic</i>	175.3	260.2	288.08	146

Notes: (1) The dependent variable is the wage growth rate as measured by the log change in wages. (2) The variable $\Delta \ln(W_t^*)$ is the wage growth rate of the base country, Peg_t is a dummy variable indicating whether a country pegs its exchange rate to its base country via a currency union or other arrangements, $\Delta \ln(a_t)$ is the productivity growth, and $CPI \text{ inflation rate}_t$ is the inflation rate as measured by the percentage change of CPI. (3) The top row indicates the time interval at which the wage growth rates are calculated. (4) The numbers in the parentheses are robust standard errors. (5) *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. (6) All regressions include country fixed effects.

Table 4: Specification 2, with productivity growth and CPI inflation

	Quarterly	Annual	2-year	4-year
Variables	(1)	(2)	(3)	(4)
$\Delta \ln(W_t^*)$	0.19 (0.08)**	0.26 (0.15)*	0.2 (0.18)	0.31 (0.19)
$CU_t \times \Delta \ln(W_t^*)$	0.4 (0.24)	0.29 (0.24)	0.61 (0.24)**	0.24 (0.27)
$Non\ CU\ peg_t \times \Delta \ln(W_t^*)$	-.16 (0.13)	0.02 (0.1)	0.18 (0.11)*	-.09 (0.17)
$\Delta \ln(a_t)$	-.05 (0.1)	0.02 (0.05)	0.03 (0.04)	0.12 (0.07)*
$CPI\ inflation\ rate_t$	0.54 (0.15)***	0.77 (0.04)***	0.81 (0.03)***	0.79 (0.09)***
<i>Linear time trend</i>	-.0001 (0.0000452)***	-.0009 (0.0006)	-.003 (0.002)*	-.01 (0.009)
<i>Const.</i>	0.02 (0.006)***	0.04 (0.02)*	0.06 (0.04)*	0.11 (0.09)
<i>Obs.</i>	1899	622	295	139
R^2	0.38	0.72	0.77	0.84
$F\ statistic$	141.62	204.29	229.19	104.09

Notes: (1) The dependent variable is the wage growth rate as measured by the log change in wages. (2) The variable $\Delta \ln(W_t^*)$ is the wage growth rate of the base country, CU_t is a dummy variable indicating membership of a currency union, $Non\ CU\ peg_t$ is a dummy variable indicating whether a country engages in a peg other than being a member of a currency union, $\Delta \ln(a_t)$ is the productivity growth, and $CPI\ inflation\ rate_t$ is the inflation rate as measured by the percentage change of CPI. (3) The top row indicates the time interval at which the wage growth rates are calculated. (4) The numbers in the parentheses are robust standard errors. (5) *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. (6) All regressions include country fixed effects.

Table 5: Specification 2, with productivity growth and CPI inflation, EMU countries only

Variables	Quarterly (1)	Annual (2)	2-year (3)	4-year (4)
$\Delta \ln(W_t^*)$	0.3 (0.08)***	0.41 (0.12)***	0.49 (0.23)**	0.48 (0.23)**
$CU_t \times \Delta \ln(W_t^*)$	0.07 (0.15)	0.09 (0.1)	0.64 (0.12)***	0.45 (0.16)***
$Non\ CU\ peg_t \times \Delta \ln(W_t^*)$	-.38 (0.08)***	-.11 (0.09)	0.12 (0.07)*	-.003 (0.06)
$\Delta \ln(a_t)$	-.02 (0.1)	0.02 (0.02)	0.08 (0.05)*	0.12 (0.06)*
$CPI\ inflation\ rate_t$	0.53 (0.13)***	0.98 (0.08)***	1.05 (0.1)***	1.08 (0.12)***
<i>Linear time trend</i>	-.0001 (0.0000254)***	-.0002 (0.0002)	0.0006 (0.001)	0.004 (0.006)
<i>Const.</i>	0.01 (0.003)***	0.002 (0.008)	-.04 (0.03)	-.08 (0.08)
<i>Obs.</i>	972	320	155	76
R^2	0.44	0.81	0.88	0.94
$F\ statistic$	91.6	96.41	158.99	753.86

Notes: (1) The dependent variable is the wage growth rate as measured by the log change in wages. (2) The variable $\Delta \ln(W_t^*)$ is the wage growth rate of the base country, CU_t is a dummy variable indicating membership of a currency union, $Non\ CU\ peg_t$ is a dummy variable indicating whether a country engages in a peg other than being a member of a currency union, $\Delta \ln(a_t)$ is the productivity growth, and $CPI\ inflation\ rate_t$ is the inflation rate as measured by the percentage change of CPI. (3) The top row indicates the time interval at which the wage growth rates are calculated. (4) The numbers in the parentheses are robust standard errors. (5) *, **, and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. (6) All regressions include country fixed effects.

Table 6: test for cointegration of nominal wages, 5% significance level

Country	Base country	Cointegration up to 1998q4	Cointegration since 1999q1	Non-currency-union pegs with base country	Wage data range
EMU members since 1999					
Austria	Germany	no	yes	1975-1998	1973q1-2008q3
Belgium	Germany	no	yes	1975-1980, 1984-1992, 1994-1998	1973q1-2010q4
Finland	Germany	no	no	1975, 1979, 1987, 1990, 1997-1998	1973q1-2010q3
France	Germany	no	no	1979-1980, 1984-1985, 1987-1994, 1996-1998	1973q1-2010q3
Germany	US	no	no		1973q1-2010q2
Ireland	Germany	no	no	1979-1980, 1984-1985, 1987-1992, 1998	1979q1-2010q3
Italy	Germany	no	yes	1980, 1984, 1986, 1988, 1990-1991, 1997-1998	1973q1-2010q4
Luxembourg	Belgium	no	no	1973-1998	1980q1-2010q3
Netherlands	Germany	no	no	1975-1976, 1978-1998	1973q1-2010q4
Portugal	Germany	NO DATA	yes		2000q1-2010q4
Spain	Germany	no	yes	1996-1998	1981q1-2010q3
Country in ERM II					
Denmark	Germany	no	yes	1975-1976, 1978, 1980, 1983-1992, 1994-2010	1973q1-2010q3
Other countries					
Australia	US	no	no		1984q1-2010q3
Canada	US	no	yes	1973-1974, 1983, 1986, 1990-1991, 1996	1973q1-2010q3
Japan	US	no	no		1973q1-2010q3
Korea	US	yes	no	1992-1994	1992q1-2010q3
Mexico	US	no	no		1980q1-2010q3
Hungary	Germany	NO DATA	no	1999	1995q1-2010q3
New Zealand	Australia	no	no	1978-1980, 1988, 1992-1993	1989q1-2010q4
Norway	Germany	no	no	1975-1996, 1984, 1990-1991, 1994	1973q1-2010q3
Poland	Germany	NO DATA	no		1995q1-2010q4
Sweden	Germany	no	yes	1974-1975, 2003-2004	1973q1-2010q3
Turkey	US	no	no		1988q1-2010q3
UK	Germany	no	no		1973q1-2010q2

Notes: (1) The test of cointegration is the Johansen cointegration test (Johansen, 1991, 1995), based on quarterly data. The critical values are obtained from MacKinnon et al. (1999) (2) "NO DATA" indicates that there is no data or insufficient data to test for cointegration. "ERM II" stands for the European Exchange Rate Mechanism II. (3) Prior to 1979, the UK is the base country for Ireland and the Irish pound had been pegged to the pound sterling. In both panel regressions and cointegration analysis, we discard the Irish data before 1979 to avoid complication.